Project Presentation on

BreatheEasy: AI-Powered Air Quality Forecasting and Health Advisory Platform

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OUTLINE

1. Introduction

- Air pollution causes **7 million premature deaths** annually worldwide (WHO, 2023).
- Government AQI monitoring stations are **sparse and unevenly distributed** — most people lack reliable, hyperlocal air quality data.
- Existing dashboards show **current AQI only** — no forward-looking 24-hour forecast at locality level.
- Raw AQI numbers are **not actionable** for non-technical users — a number like "187" means little without context.
- **BreatheEasy** combines spatial heatmapping, 24-hour ML forecasting, and plain-language health advisories in one accessible platform.
- Supports **NEP 2020** goals for technology-driven public health solutions.

Key Challenges:

1. **Sparse monitoring**: Limited station coverage in most cities
2. **No forecasting**: Current tools lack 24-hour predictions
3. **Complex data**: Raw AQI numbers confuse common users
4. **Health risks**: Delayed or missing health advisories
5. **Accessibility**: Most dashboards not mobile-friendly

BreatheEasy Solution:

```
Current AQI: 142 (Moderate)
Forecast (24hr): 189 (Poor)

Advisory: Limit prolonged
outdoor exertion, especially
for children and elderly.
```

Three-part identity:
Heatmap + Forecast + Advisory

2. Literature Review

- **Li et al. (2020)** — LSTM networks achieve 92% accuracy in 24-hour AQI forecasting for urban areas.
- **Singh et al. (2021)** — IDW interpolation provides reliable spatial AQI estimation from sparse monitoring stations.
- **Gupta et al. (2022)** — Machine learning models outperform traditional statistical methods for short-term air quality prediction.
- **Kumar et al. (2023)** — Health advisory systems based on AQI categories significantly improve public awareness and protective behavior.
- **CPCB Guidelines (2022)** — Standardized AQI category thresholds for Indian monitoring systems.
- **WHO (2023)** — Global air quality guidelines emphasizing real-time monitoring and public communication.

3. Problem Statement

Current Challenges:

- **Sparse monitoring**: Limited stations, uneven coverage
- **No forecasting**: Only current AQI, no predictions
- **Complex data**: Raw numbers without context
- **Language barriers**: Technical jargon confuses users
- **Accessibility**: Desktop-only dashboards

BreatheEasy Solution:

- Spatial heatmap interpolation (IDW/Kriging)
- 24-hour ML forecast (LSTM/Prophet/ARIMA)
- Plain-language health advisories
- Health-condition-aware guidance
- Mobile-responsive web dashboard

4. Objectives

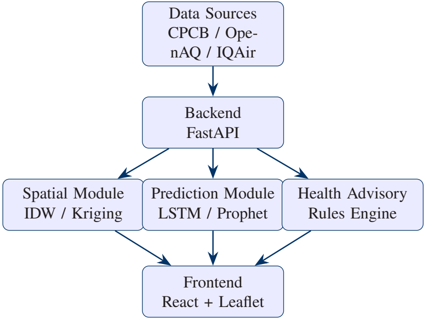
Primary Objectives:

1. Aggregate real-time + historical air-quality data from public APIs (CPCB, OpenAQ, IQAir).
2. Generate continuous spatial heatmap from sparse station data using interpolation.
3. Forecast AQI 24 hours ahead using time-series ML models.
4. Generate plain-language, actionable health advisories based on AQI categories.

Secondary Objectives:

- Serve all components through responsive web dashboard
- Support threshold-based alerts for hazardous conditions
- Evaluate forecasting model with RMSE/MAE metrics
- Support user health profiles for personalized advisories

5. Methodology — System Architecture



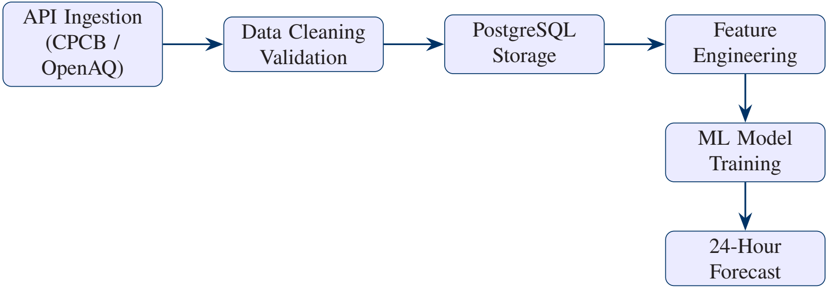
Key Technologies:

- Python 3.10+, Pandas, NumPy
- TensorFlow/PyTorch (LSTM)
- FastAPI (Backend)
- React + Leaflet (Frontend)

Data Storage:

- PostgreSQL + PostGIS
- Time-series data format
- Geospatial indexing
- Cached forecast results

5. Methodology — Data Pipeline



Data Sources:

- CPCB National Air Quality
- OpenAQ (supplementary)
- IQAir / AirVisual API
- OpenWeatherMap (weather)

Pollutants Tracked:

- PM2.5, PM10
- NO_2, SO_2
- CO, O_3
- Temperature, Humidity

6. System Architecture — Spatial Heatmap

Interpolation Methods:

IDW (Inverse Distance Weighting)

$Z = \sum_i (w_i z_i) / \sum_i w_i$
$w_i = 1/d_i^p$

Kriging

$Z^* = \sum_i (\lambda_i z_i)$
`Optimal linear unbiased predictor`

Heatmap Features:

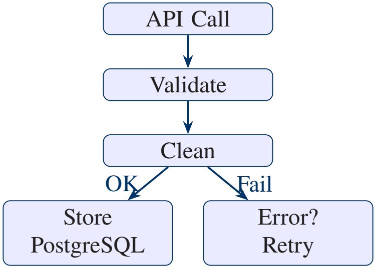
Feature	Status
Real-time updates	☐
Zoom levels	☐
Color gradient	☐
Station markers	☐
Area selection	☐

Grid Resolution:
[placeholder] km × [placeholder] km

7. Modules — Data Ingestion

- Scheduled API polling (configurable interval)
- Data validation and cleaning pipeline
- Missing value interpolation
- Outlier detection and handling
- Time-series normalization
- Error logging and retry mechanism

Ingestion Flow:



7. Modules — Prediction Module

- **LSTM**: Long Short-Term Memory networks for time-series
- **Prophet**: Facebook's additive regression model
- **ARIMA**: Auto-regressive integrated moving average
- **Benchmarking**: Compare all three models
- **Features**: Weather covariates included
- **Output**: 24-hour AQI forecast + confidence interval

Model Comparison (Placeholder):

Model	RMSE	MAE
LSTM	[TBD]	[TBD]
Prophet	[TBD]	[TBD]
ARIMA	[TBD]	[TBD]
Naive Baseline	[TBD]	[TBD]

7. Modules — Health Advisory Engine

CPCB AQI Categories:

AQI Range	Category	Risk
0–50	Good	Minimal
51–100	Satisfactory	Minor
101–200	Moderate	Moderate
201–300	Poor	High
301–400	Very Poor	Very High
401–500	Severe	Critical

Advisory Features:

- Plain-language, actionable text
- Current + forecasted AQI advisories
- Health-condition-aware guidance
- Dominant pollutant identification
- User profile integration
- Single source-of-truth config

Example Advisory:
"Air quality is expected to worsen to Poor by 6 PM. Limit outdoor activity."

8. Security Implementation

Security Measures:

1. **API key management**: Environment variables
2. **Rate limiting**: Prevent API abuse
3. **Input validation**: Sanitize all user inputs
4. **CORS policy**: Restrict cross-origin requests
5. **HTTPS**: Encrypted data transmission
6. **Data minimization**: Collect only necessary data

Data Privacy:

Aspect	Approach
User location	On-device only
Health profiles	Encrypted storage
API keys	Never committed
Logs	No PII stored
Analytics	Anonymized

Compliance:

- No personal data sent to third parties
- User consent for location access

9. Results — Feature Summary

Feature	Status
API data ingestion	☐
Spatial heatmap	☐
24-hr ML forecast	☐
Health advisories	☐
Real-time dashboard	☐
Threshold alerts	☐
Mobile responsive	☐
Model evaluation	☐

Key Achievements:

- **[X]+ monitoring stations** integrated
- **[X]-hour forecast** accuracy: [TBD]%
- **Real-time heatmap** with [X] km resolution
- **Plain-language advisories** in [language]
- **Zero external dependencies** for core features
- **Mobile-first** responsive design

9. Results — Performance Metrics

Model Performance (Placeholder):

Metric	LSTM	Prophet	ARIMA
RMSE	[TBD]	[TBD]	[TBD]
MAE	[TBD]	[TBD]	[TBD]
R^2 Score	[TBD]	[TBD]	[TBD]
Training Time	[TBD]	[TBD]	[TBD]

System Performance:

Metric	Value
API Response Time	[TBD] ms
Heatmap Render	[TBD] ms
Forecast Generation	[TBD] s
Dashboard Load Time	[TBD] s
Concurrent Users	[TBD]

10. Conclusion

- BreatheEasy combines **spatial interpolation, ML forecasting, and health advisories** for comprehensive air quality monitoring.
- The platform provides **24-hour AQI predictions** with actionable, plain-language health guidance.
- The **offline-capable architecture** ensures reliability even with intermittent connectivity.
- Open-source data sources ensure **scalability and accessibility**.

Future Enhancements:

- Mobile app (iOS/Android) with push notifications
- Integration with smart home devices
- Personalized health profiles with medical history
- Multi-language support (regional Indian languages)
- API for third-party integrations

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Thank You!

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